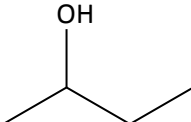
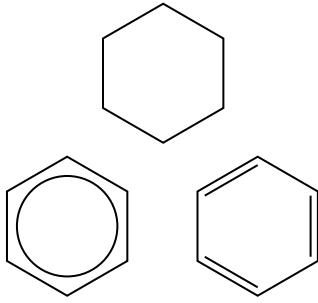


4.1.1 Basic concepts of organic chemistry

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Naming and representing the formulae of organic compounds	
(a) application of IUPAC rules of nomenclature for systematically naming organic compounds	Nomenclature will be limited to the functional groups within this specification. E.g. $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$ has the systematic name: 2-methylbutan-1-ol. Learners will be expected to know the names of the first ten members of the alkanes homologous series and their corresponding alkyl groups. HSW8 Use of systematic nomenclature to avoid ambiguity. HSW11 The role of IUPAC in developing a systematic framework for chemical nomenclature.
(b) interpretation and use of the terms: (i) <i>general formula</i> (the simplest algebraic formula of a member of a homologous series) e.g. for an alkane: $\text{C}_n\text{H}_{2n+2}$ (ii) <i>structural formula</i> (the minimal detail that shows the arrangement of atoms in a molecule) e.g. for butane: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ or $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$ (iii) <i>displayed formula</i> (the relative positioning of atoms and the bonds between them) e.g. for ethanol: $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ (iv) <i>skeletal formula</i> (the simplified organic formula, shown by removing hydrogen atoms from alkyl chains, leaving just a carbon skeleton and associated functional groups) e.g. for butan-2-ol: 	<i>M4.2</i> See also 2.1.3 b for empirical formula and molecular formula. Definitions not required. In structural formulae, the carboxyl group will be represented as COOH and the ester group as COO . The symbols below will be used for cyclohexane and benzene:  HSW8 Communication using organic chemical structures; selecting the appropriate type of formula for the context.

Functional groups

- (c) interpretation and use of the terms:
- (i) *homologous series* (a series of organic compounds having the same functional group but with each successive member differing by CH_2)
 - (ii) *functional group* (a group of atoms responsible for the characteristic reactions of a compound)
 - (iii) *alkyl group* (of formula $\text{C}_n\text{H}_{2n+1}$)
 - (iv) *aliphatic* (a compound containing carbon and hydrogen joined together in straight chains, branched chains or non-aromatic rings)
 - (v) *alicyclic* (an aliphatic compound arranged in non-aromatic rings with or without side chains)
 - (vi) *aromatic* (a compound containing a benzene ring)
 - (vii) *saturated* (single carbon–carbon bonds only) and *unsaturated* (the presence of multiple carbon–carbon bonds, including $\text{C}=\text{C}$, $\text{C}\equiv\text{C}$ and aromatic rings)
- (d) use of the general formula of a homologous series to predict the formula of any member of the series

Definition required for homologous series only.

R may be used to represent alkyl groups, but also other fragments of organic compounds not involved in reactions.

The terms saturated and unsaturated will be used to indicate the presence of multiple carbon–carbon bonds as distinct from the wider term ‘degree of saturation’ used also for any multiple bonds and cyclic compounds.

Isomerism

- (e) explanation of the term *structural isomers* (compounds with the same molecular formula but different structural formulae) and determination of possible structural formulae of an organic molecule, given its molecular formula

M4.2

Reaction mechanisms

- (f) the different types of covalent bond fission:
- (i) homolytic fission (in terms of each bonding atom receiving one electron from the bonded pair, forming two radicals)
 - (ii) heterolytic fission (in terms of one bonding atom receiving both electrons from the bonded pair)

(g) the term *radical* (a species with an unpaired electron) and use of 'dots' to represent species that are radicals in mechanisms

(h) a 'curly arrow' described as the movement of an electron pair, showing either heterolytic fission or formation of a covalent bond

(i) reaction mechanisms, using diagrams, to show clearly the movement of an electron pair with 'curly arrows' and relevant dipoles.

Radical mechanisms will be represented by a sequence of equations.

Dots, •, are required in all instances where there is a single unpaired electron (e.g. $\text{Cl}\bullet$ and $\text{CH}_3\bullet$). Dots are **not** required for species that are diradicals (e.g. O).

'Half curly arrows' are **not** required, **see 4.1.2 f.**

HSW1,8 Use of the 'curly arrow' model to demonstrate electron flow in organic reactions.

Any relevant dipoles should be included.

Curly arrows should start from a bond, a lone pair of electrons or a negative charge.

HSW1,2,8 Use of reaction mechanisms to explain organic reactions.